Image Caption Generation in Kannada using Deep Learning Frameworks

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***Abstract*—The official language of the state of Karnataka and one of the most frequently spoken languages in India, Kannada is spoken by more than 57 million people worldwide. In this modern era, one of the most important tasks in the field of Natural Language Processing in the present era is the development of image captions. In order to provide a textual description of an image intended to highlight its key qualities, Image Caption generation has long been a study of significant interest to academics in the fields of Artificial Intelligence and Machine Learning. While a great deal of research is being conducted in English, resources in regional languages like Kannada, are severely lacking. Given that 72% of the Karnataka population are Kannada speakers, Image Caption generation in Kannada is the realization of human-computer interaction for native language speakers, enabling the development of regional language virtual assistants which is a breakthrough in this line of work. Through our research, we present a model that generates semantic, syntactic and structured captions in Kannada.**

***Index Terms*—Encoder-Decoder architecture, Convolution Neural Network, Recurrent Neural Network, Image Captioning, RESNET50, Vision Transformers, VGG-16, Flickr8K**

I. Introduction

Image Caption generation involves discerning important objects, their attributes, and the relationships among the objects in the image to generate syntactically and semantically accurate descriptions. This is an eager endeavor to further Image Caption generation in Kannada and thus pave the path for further research and enhancement in this discipline. This study describes a system that creates image captions using Encoder-Decoder architecture. In our study, we first extract the image's features using a pre-trained Convolutional Neural Network (CNN) called VGG-16 and RESNET-50, and then we create a contextually relevant caption using a Recurrent Neural Network (RNN). This paper also introduces a new and novel architecture for Image Caption generation in Kannada using a transformer encoder. Recurrent neural networks and convolutional neural networks are used in this model to encode the variable-length input into a fixed-dimensional vector and then "decode" it into the desired output caption. The architecture consists of a transformer encoder, patch and positional embedding, RNN and a multi-head attention model. The concept of attention introduces another level of innovation and novelty to our project.

There is an abundance of datasets available for Image Caption generation in the English language, including Flickr30K, MSCOCO, Flickr8K etc. The Flickr30k dataset is a sought-after benchmark for Image Caption generation. The dataset has over 31,000 images, and each of these images have five captions that describe the varied contents of the image. The notations on each of these images allow for advancement in automatic image description and grounded language interpretation. We tested our model on the comparatively smaller Flickr8K dataset which consists of 8091 images with five captions for each image that describe the image in diverse ways.

II. Literature Survey

A. Ramesh G, Atul M Bharadwaj, Srihari N, Champa HN UVCE "Kannada ImageNet: A Dataset for Image Classification in Kannada", 2021

This paper talks about developing a Kannada dataset for Image Classification by using a sample of 1000 images with the help of the ImageNet Dataset that is generally used

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for classification in the English language. The dataset was classified into 890 classes initially, 193 classes were later added manually after cross verifying it with the WordNet synsets already available. As there is no official information regarding the accuracy of the translations done from the English language to Kannada and hence a large portion of the work was done manually. The translated labels were categorized into "Accurate", "Inaccurate", "Neutral" or "English" where "Accurate" and "English" are grouped as accurate and "Inaccurate" and "Neutral" are grouped as inaccurate labels. A human annotator was used to check the correctness of the in the Kannada Language. Google Translate API was used to translate the labels and the total translation accuracy observed was 97.15%.

B. Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio "Generating Image Descriptions using Multilingual Data", 2017

The paper explored several models which attempt to generate captions for both languages, ignoring the English output during evaluation. Comparison of the results to a baseline implementation which uses only the German captions for training shows significant improvement in the model proposed. Usage of LSTM with attention over both the English embeddings and the image produces a German caption was introduced in the paper.

C. Santosh Kumar Mishra, Rijul Dhir, Sriparna Saha, and Pushpak Bhattacharya, Indian Institute of Technology, Patna "A Hindi Image Caption Generation Framework Using Deep Learning", 2021

This study suggested creating a Hindi-language Image Caption generation model utilizing an encoder-decoder architecture since it increases the likelihood of producing an accurate caption given an image. GRU served as the decoder with spatial attention, and RESNET101 served as the encoder. The popular MSCOCO dataset was manually translated from English to Hindi to create a dataset.

There are various attention-based architectures that have been created for Hindi image caption generating. The generated results of the suggested model were evaluated by comparing their BLEU scores to a number of baselines.

D. Mert Kilickaya, Aykut Erdem, Nazli Ikizler-Cinbis, and Erkut Erdem Hacettepe University Computer Vision Lab Dept. of Computer Engineering, Hacettepe University, Ankara, Turkey "Re-evaluating Automatic Metrics for Image Captioning", 2017

This paper talks about how a number of human-written reference (gold standard) descriptions and an automatically generated caption are compared in order to determine how similar or dissimilar they are. It also provides an emphasis on Word Mover's Distance (WMD) metric as this is advantageous over the other metrics.

It furnishes a brief description of the metrics BLEU, ROUGE, WMD, CIDEr, METEOR, SPICE along with the methods to calculate them. The paper also describes the challenges and drawbacks of the various metrics in use presently. Moreover, the paper discusses the effect of evaluation metrics on quality, accuracy and robustness using datasets such as MSCOCO, Flickr8K, Flickr30K. The conclusion bestowed is that a combination of WMD, SPICE, METEOR is generally the best.

E. Kelvin Xu, Jimmy LeBa, Ryan Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhutdinov, Richard S. Zemel, Yoshua Bengio "Show Attend Tell: Neural Image Caption Generation with Visual Attention", 2016

This paper suggests an attention-based model to describe the contents of an image. Two attention based caption generators that is, soft and hard were introduced. CNN was used as an encoder with LSTM being used as a decoder.

The paper also encouraged the use of soft and hard models that can make the model more robust. Drawback of the proposed approach is that there is no major performance boost that was observed using ensemble comparison instead of the single method which is usually not the observed trend.

F. Dr. Vinayak D. Shinde, Mahiman P. Dave, Anuj M. Singh, Amit C. Dubey Head of Department of Computer Engineering, Shree L.R. Tiwari College of Engineering, Maharashtra, India, B.E. Student, Computer Engineering, Shree L.R. Tiwari College of Engineering, Maharashtra, India "Image Caption Generator using Big Data and Machine Learning", 2020

This paper proposed a methodology for caption generation in English using CNN and LSTM based on an encoder-decoder architecture. The generation of dense feature vector for input image which generates the caption for the given image was mentioned in the paper.

A brief description and usage of the functional modules such as CNN, LSTM and an explanation on data exploration and proposed methodology was also included. The conclusion inferred is that a larger dataset provides better accuracy and model cannot predict words outside its vocabulary.

III. DATASET

Our work employs the Flickr8K dataset for the training of our model. Due to the lack of availability of Kannada language Image Caption generating datasets, a Python script was used for the translation of the captions available in the English Image Caption generating datasets. The Flickr8K dataset contains 8091 images with five captions per each image.

After translation, the Flickr8K Kannada dataset has 3,49,934 words and 29,99,469 characters with a maximum number of captions having an average caption length of 8. The minimum and maximum number of words per caption are 5 and 20. The dataset was then checked for linguistic errors and those errors were rectified manually. The main linguistic errors were the literal translation of words from English to the Kannada language.



Fig. 1. Corresponding Image for the captions from the Flickr8K dataset

ENGLISH CAPTIONS	CORRECTED KANNADA CAPTIONS
A child in a pink dress is climbing up a set of stairs in an entryway.	ಗುಲಾಬಿ ಬಣ್ಣದ ಡ್ರೆಸ್ ತೊಟ್ಟ ಮಗುವೊಂದು ಪ್ರವೇಶ ಮಾರ್ಗದಲ್ಲಿ ಮೆಟ್ಟಿಲುಗಳನ್ನು ಹತ್ತುತ್ತಿದೆ.
A girl going into a wooden building.	ಮರದ ಕಟ್ಟಡಕ್ಕೆ ಹೋಗುವ ಹುಡುಗಿ.
A little girl climbing into a wooden playhouse.	ಒಂದು ಚಿಕ್ಕ ಹುಡುಗಿ ಮರದ ಪ್ಲೇಹೌಸ್‌ಗೆ ಏರುತ್ತಾಳೆ.
A little girl climbing the stairs to her playhouse.	ಪುಟ್ಟ ಹುಡುಗಿ ತನ್ನ ಆಟದ ಮನೆಗೆ ಮೆಟ್ಟಿಲುಗಳನ್ನು ಹತ್ತುತ್ತಾಳೆ.
A little girl in a pink dress going into a wooden cabin.	ಗುಲಾಬಿ ಬಣ್ಣದ ಉಡುಪಿನಲ್ಲಿ ಮರದ ಕ್ಯಾಬಿನ್‌ಗೆ ಹೋಗುತ್ತಿರುವ ಪುಟ್ಟ ಹುಡುಗಿ.

Table. 1. English Captions with the corresponding Corrected Kannada Captions.

IV. IMPLEMENTATION

A. Overview

This section is primarily about designing the architecture of the model, the implementation of the model, the usage of relevant tools, and the latest research pertaining to the different aspects of this research paper. The implementation so far consists of preparation of the Kannada dataset (Flickr8K), experimentation with various Encoder architectures and evaluation of the final output captions. In machine translation, inputs and outputs are of unaligned and varying lengths. The approach to handle this issue is straightforward. The solution is to design an Encoder-Decoder architecture. Our design contains two important components: first, an Encoder that receives a varying-length sequence as the input, and second, a Decoder that functions as a conditional language model by taking in the encoded input and the leftwards context of the target sentence and predicting the successive token in the target sentence. Recurrent neural networks and convolutional neural networks are used in this model to encode the variable-length input into a fixed-dimensional vector and then "decode" it into the desired output caption. The Encoders used are two different types of CNN – VGG –16 (with 16 convolution layers) and RESNET – 50 (with 50 convolution layers) and a type of transformer – Vision transformer. The Decoder used for all three encoders is RNN (Recurrent Neural Network). Input images and captions have been based off of the Flickr8K dataset. The average length of the captions is 8 words and the length of the captions vary between 5 and 15 words. The input image is of the size 224x224x3 for VGG–16, RESNET – 50 and 229x229x3 for Vision transformer.

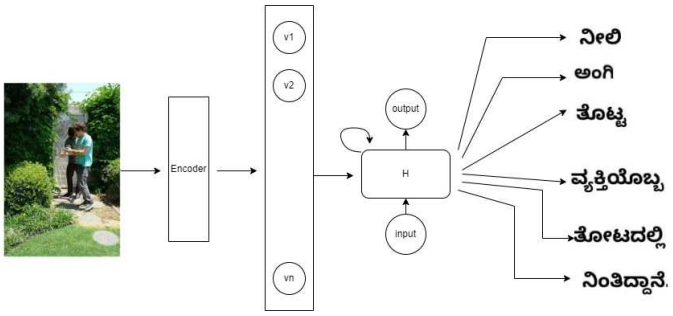


Fig. 2. Encoder - decoder architecture for image captioning in Kannada.

B. Translating the captions from English to Kannada using Google Translate API

The English captions were translated using the Google Cloud Translate API. This particular API can be used for the translation of large amounts of given input text from one language to the selected output language. The accuracy of the translations from English to regional languages like Kannada are not known but it is known that English to Arabic is the most accurate. We have used Google Translate API instead of other translation services like Bing, IBM Yandex, etc. due to its reliability and ease of access.

C. Human annotation and correction on the translated captions

The translated captions were cross checked to make sure the translated captions are up to the mark and can be used as train and test data for the image caption generation model. Some of the captions contain English words that direct translations due to limitations in the Kannada Dictionary. For example: The English word "Laser" does not have a Kannada counterpart and thus, is translated directly.

D. Resizing all the images to the same size in pixels

The Encoder-Decoder architecture used requires us to have all the images of the same size - i.e., of the same dimensions. It was made sure that all the images in the dataset were of the same size.

E. Encoder architecture

1) VGG - 16: In the 2014 ILSVRC competition, VGG-16 was among the architectures with a very high performance.

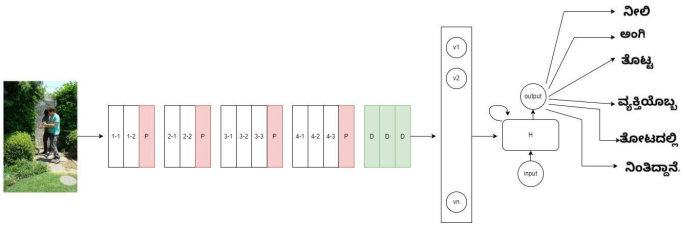


Fig. 3. Network architecture for VGG-16.

With a top-5 classification error of 7.32%, it came in second place behind GoogleNet (with a classification error of 6.66%) in the classification task. With a localization error rate of 25.32 percent, it also won the localization challenge.

Input to VGG-16 is of the size 224x224x3. The top two layers have the same padding with 64 channels of size 3*3 filters. The next layer is a max pool. The two layers after max pool have convolution layers of filter size 128. These two layers are followed by a layer of max pool. The next two convolution layers have 256 filters of size 3*3. The following layer sets each have 512 filters of size 3*3 and 3 convolution layers and 1 max pool layer. To prevent the spatial feature of the image, 1-pixel padding is applied after each convolution layer. The output from these layers is a feature map which is flattened to make a feature vector. The last three dense layers are fully connected and the third layer is used to implement a softmax function.

Long training times and exploding gradients problem are the limitations of VGG-16.

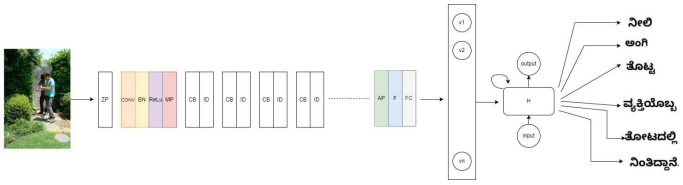


Fig. 4. Network architecture for RESNET - 50.

2) RESNET - 50: The RESNET-50 model is a popular deep residual network which is 50 layers deep. It is an ANN which uses residual blocks one over the other in a stack like formation to form a network. The model won the first place at the "ILSVRC 2015 Classification Competition" with a diminutive error rate 3.57% and was thus exceedingly successful.

RESNET models are used to tackle the problem of vanishing or exploding gradients. Usage of residual blocks improves the accuracy of models. The core principle of residual blocks is skipping connections. Skip connections work in two ways. They provide an alternative shortcut for the gradient to pass through thereby tackling the problem of vanishing and exploding gradients. The performance of the higher layers of the model are not worse when compared to the lower layers as the model learns the identity function. Basically, a skip connection adds the output from the previous layer to the output from the stacked layer. The RESNET – 50 architecture utilizes a 3-layer bottleneck block. One advantage of RESNET is that it has lower complexity and a fewer number of filters. The design of the architecture is simple - the layers had the same number of filters for the same output feature map size, and the number of filters doubled in case the feature map size was halved in order to preserve the time complexity per layer.

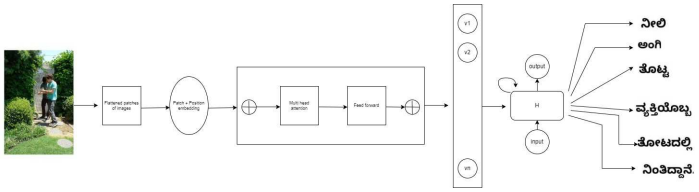


Fig. 5. Network architecture for Vision Transformer.

3) Vision Transformers: The Encoder-Decoder architecture used by RNN and the vision transformer network is similar. The ability to feed the input to the transformers in parallel is the main distinction. This shows that the input is not connected to a time step. The complete English sentence is provided to the input embedding while translating from English to Kannada. Images that represent a position in space where words with similar meanings are physically closer to one another can serve as the input embedding layer. Each word is mapped to a vector of continuous values in order to represent it.

When this is done, there is an issue in that the same word can have various meanings in different phrases. Thus, position encoding is employed to address this problem. Transformers lack recurrence and convolution. The model must therefore make use of information about the relative or absolute positions of words in a phrase in order to use the sequence's order.The final concept is using predetermined or learned weights to encrypt data relevant to a particular token's location in a sequence. Likewise, the target Kannada word and its positional encoding vector are sent to the output embedding. The decoder block receives these next.

The encoder block is divided into two levels. A simple, positionally fully connected feed-forward network makes up the second sub-layer, whereas the first sub-layer is a multi-head self-attention mechanism. In order to identify contextual and pertinent associations between words in a phrase, each word has an attention vector that is generated. The encoder uses a particular attention mechanism called self-attention for multi-headed attention. Each token in the input sequence can be connected to other tokens thanks to the models' ability to pay attention to themselves. In addition to the two sub-layers in each encoder layer, the decoder feeds a third sub-layer which uses multi-head attention over the output of the encoder stack. Following layer normalization, residual connections are implemented around each of the sub-layers. The encoder sends the attention vectors for the English sentences and Kannada words to the second multi-head attention.

This attention block will now determine the relationships between each pertinent word vector. Here is where the term

from English is translated into Kannada. A linear layer that serves as a classifier and a softmax are used by the decoder to determine the word probabilities.

4) Extracting features from the Images: We experimented with a variety of deep CNNs like VGG-16, RESNET-50 and with transformer architecture such as Vision transformer to extract relevant features from the image dataset. The output of the last layer of CNNs are used in the process of Image classification rather than just feature extraction. The output of the last layer was discarded, and the output of the pre-final layer was stored, since what we are interested in is feature extraction.

5) Decoder architecture - RNN: As the captions generated must be coherent, there is a need to remember the previously predicted word in the caption. Hence, RNN or Recurrent Neural Network, a type of neural network is used to predict the next word in the sentence. In RNNs, the output from the previous step is fed as input to current step. The most important feature of RNNs is the hidden state. RNNs have a "memory" which remembers all information about what has been calculated. RNNs make use of the same parameters for each input as the same task is performed on all the inputs or hidden layers to produce the output. This complexity of parameters is hence reduced.

6) Evaluation metric – BLEU score: BLEU score is Bilingual Evaluation Understudy. It is an evaluation metric that is used to assess the quality of captions that have been generated through the ML model. BLEU is language independent, easy to compute and understand. The BLEU score lies between 0 and 1. A BLEU score between 0.4 and 0.5 implies high quality machine translations. A BLEU score of 0.6 is considered to be the highest score that can be achieved.

For calculation of BLEU score, two concepts are used: n-grams, precision. The precision for 1-gram to 4-grams is first calculated for each caption. The geometric average precision scores are then calculated with N = 4 and w_n = N/4. The formula is as follows:

$$N = exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \quad (1)$$

Brevity penalty is then calculated. Finally, the BLEU score is computed by multiplying the Geometric Average Precision with the Brevity Penalty. Limitations of BLEU score are that it does not consider the meaning of words or the word order. It looks only for exact word matches. It also ignores the importance of words.

V. RESULTS AND DISCUSSION

Utilization of transformer encoder and RESNET for the pipeline have both yielded satisfactory results. While RESNET performs better for the entire dataset, Vision Transformers work well for the individual datasets. Since, the BLEU score is not context inclusive, especially for regional languages, this evaluation metric cannot be determined to be the only metric for the evaluation of the Image Caption Generation model. Other performance metrics can also be deployed to check the accuracy of the predicted captions. While generating the captions for Vision Transformers we noticed that even though the predicted captions were correct and coherent, the BLEU score for some images was 0.0 as the unigram values in the actual and predicted captions did not match, whereas, the BLEU scores for some images were exceptionally good. A few images had a BLEU score of 0.528 to 0.60 indicating the generation of highly accurate descriptions. As seen in the figure below, the corpus BLEU performance values for VGG-16 is 0.36, Vision Transformers in 0.41 and for RESNET is 0.436. The generated outputs are valid descriptions of the images. They are syntactically and grammatically correct captions which fit the context of the image and they also recognize multiple objects within the image. In order to generate much higher quality captions, the training hours have to be increased substantially. In order to counter images which are originally blurry, introduction of noise in the dataset during model training will prove helpful.

ENCODER ARCHITECTURE	CORPUS BLEU SCORES USING FLICKR8K
RESNET50	0.436
Vision Transformer	0.41
VGG-16	0.36

Table. 2. Comparison of BLEU scores for the various encoders.

VI. CONCLUSION AND FUTURE WORK

Image caption generation in Kannada is a novel research study. This project was centered on the generation of valid descriptions for images in the Kannada language. The captions had to be grammatically, syntactically coherent and correct sentences. From the preparation of the dataset using the existing Flickr8K dataset in English to the final caption generation, numerous architectures and deep learning models were explored and used.

The datasets were prepared to be used in the Kannada language using the Google Translate API. Manual corrections were required to be done for the generated captions to ensure that the model was trained on an impeccable dataset withclose to no flaws. The architecture of the model is an Encoder–Decoder architecture and with each change in the encoder architecture, we recorded a better performance of the model. There was a significant improvement in the BLEU scores as we implemented VGG-16 and then RESNET-50 and then finally Vision Transformers. Further, improved accuracy can be obtained by implementing a hybrid model.

In conclusion, the proposed algorithm and model is the foundation work for further research that will aid millions of regional language and native speakers utilize the latest technologies in the easiest way available to them. Furthermore, our work, will support the development of regional language virtual assistants that will in turn be the ground-breaking advancement of technology in this era. This model can also be used to help visually impaired people make use of technology better, thereby enhancing the human computer interaction further. Rolling out text-to-speech models in regional languages will truly be a boon to the visually impaired persons in the society.

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REFERENCES

[1] Dr. Vinayak D Shinde, Mahiman P. Dave, Anuj M Singh, Amit C Dubey Tiwari College of Engineering, Mumbai "Image Caption Generator using Big data and Machine Learning"

[2] Vinay Uday Prabhu. "Kannada-MNIST: A new handwritten digits' dataset for the Kannada language" 2019 arXiv: 1908.01242[cs.CV]

[3] Mert Kilickaya, Aykut Erdem, Nazli Ikizler-Cinbis, and Erkut Erdem Hacettepe University Computer Vision Lab Dept. of Computer Engineering, Hacettepe University, Ankara, TURKEY. "Re-evaluating Automatic Metrics for Image Captioning"

[4] Santosh Kumar Mishra, Rijul Dhir, Sriparna Saha, Pushpak Bhattacharyya, IIT, Patna, "A Hindi Image Caption Generation Framework Using Deep Learning"

[5] Kishore Papineni, Salim Roukos, ToddWard, andWei- Jing Zhu. 2002. "BLEU: a method for automatic evaluation of machine translation". In ACL, pages 311–318

[6] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. 2014. Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. "Generating Image Descriptions using Multilingual Data"

[7] Kelvin Xu, Jimmy LeBa, Ryan Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhutdinov "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention"

[8] Wenpeng Yin, Katharina Kann, Mo Yu, and Hinrich Schütze. 2017. Comparative study of CNN and RNN for natural language processing. arXiv preprint arXiv:1702.01923 (2017).

[9] Raimonda Staniut ̄ e and Dmitrij Šešok. 2019. A systematic literature review on image captioning. ˙ Appl. Sci. 9, 10 (2019), 2024.

[10] n. neurohive. VGG16-convolutional network for classification and detection. Internet: https://neurohive.io/en/popular-networks/vgg16/, [Sep. 13, 2019].

[11] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," arXiv preprint arXiv:1409.1556, 2014

[12] X. Chen and C. Lawrence Zitnick, "Mind's eye: A recurrent visual representation for image caption generation," in Proceedings of the IEEE conference on computer vision and pattern recognition, 2015, pp. 2422–2431.

[13] Oriol Vinyals, Alexander Toshev, Samy Bengio, and Dumitru Erhan. 2015. Show and tell: A neural image caption generator. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 3156–3164.

[14] Aditya Deshpande, Jyoti Aneja, Liwei Wang, Alexander G. Schwing, and David Forsyth. 2019. Fast, diverse and accurate image captioning guided by part-of-speech. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR'19).

[15] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in Advances in neural information processing systems, 2012, pp. 1097–1105.

[16] Tsung-Yi Lin et al. "Microsoft COCO: Common Objects in Context". In: Computer Vision – ECCV 2014. Ed. by David Fleet et al. Cham: Springer International Publishing, 2014, pp. 740–755